

Automated Tea/Coffee Cooling

Carter Norsen and Evan Hill

It is important to note that this idea is not final, as our group has not yet been able to communicate to discuss ideas and the idea chosen may not fully meet project requirements, as they are yet to be published. However, the current idea was chosen to be adaptable to future complexity requirements, so it may be a valid idea.

ABSTRACT

Although not usually problematic, it is a common occurrence to prepare a cup of tea or coffee using high-temperature water and leave it alone to cool for a while, either for too long or too short of a time so the desired temperature is never reached. While this could be remedied with a timer, careful experimentation is required by every user and environment to achieve the desired temperature, so an automated method could be better. This project will be centered around the DHT11 temperature and humidity sensor to determine the temperature of the tea or coffee, while the logic could be controlled by an UNO R3 controller or by digital logic using more basic components, depending on project requirements, but either method could be effective. The other part of the design includes a method of alerting the user that the tea or coffee is ready, which has not been decided yet. Some ideas for that include combinations of a speaker, a light display, or a separate circuit that alerts the user at their location. Additional complexity could result from the necessity of a fan to cool the tea or coffee and improve the reading of the DHT11 sensor, a structure to arrange the components of the design, and the potential need for temperature calibration. The project does not require much funding, as all DU Electrical Engineering students already own an UNO R3 controller, a DHT11 sensor, a fan and motor, some basic logic circuitry, and some wires, so additional parts required should not be difficult to procure, aided by the fact that the user could be alerted in a multitude of creative and fun ways.

MEMBER ROLES

Again, no communication has been successful, so these are temporary divisions of labor and subject to change.

Carter Norsen: Design of logical circuit, assembly, procurement of additional components, provision of tea, documentation.

Evan Hill: Design of the logical circuit, assembly, documentation.